TASK 1: Repository Setup

Goal: Initialize Git and create first commit

Steps:

1. Create a folder named git-assignment-yourfullname

2. Navigate into the folder

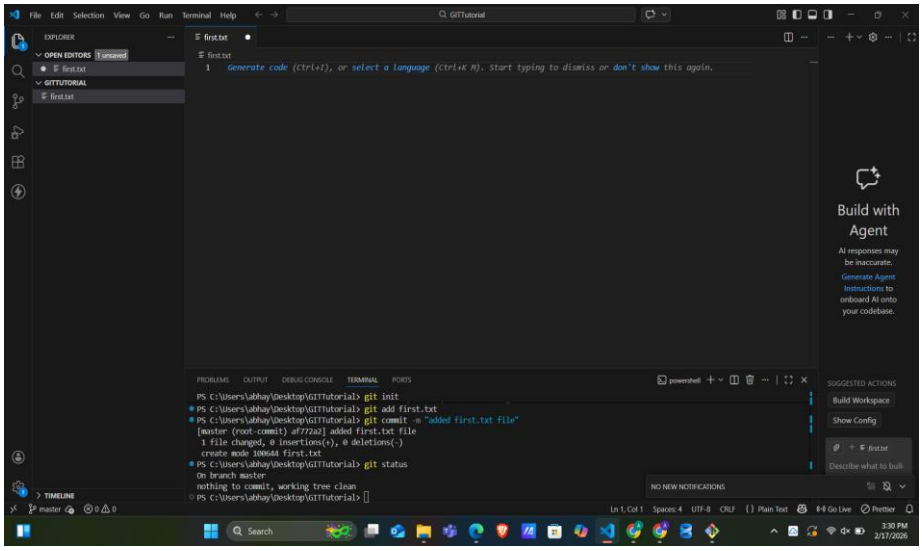
3. Initialize Git repository

4. Create a file named index.txt

5. Check repository status

6. Stage the file

7. Commit with message: Initial project setup



TASK 2: Branch Creation & Work

Goal: Work independently in a branch

Steps:

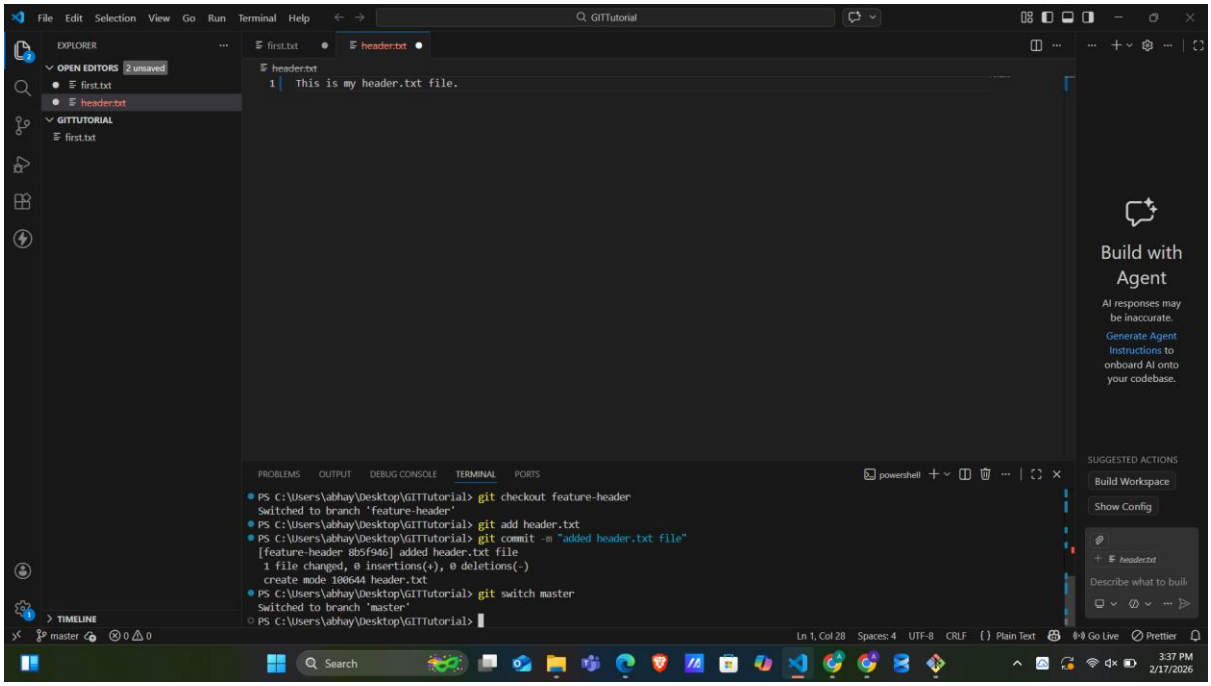
1. Create a branch named feature-header

2. Switch to feature-header

3. Create a file named header.txt

4. Stage the file

5. Commit with message: Added header section



TASK 3: Branch Switching Observation

Goal: Observe branch isolation

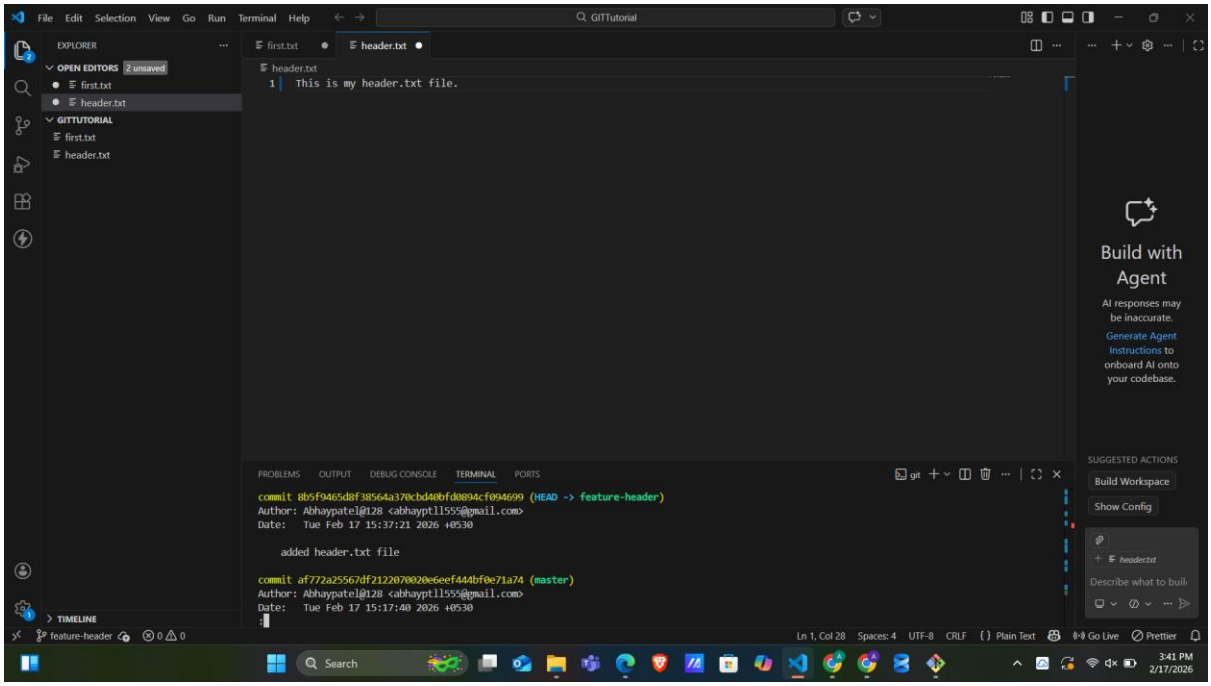
Steps:

1. Switch back to master

2. List files and observe

3. Switch again to feature-header

4. List files and observe



TASK 4: Undo a Mistake

Goal: Safely undo file changes

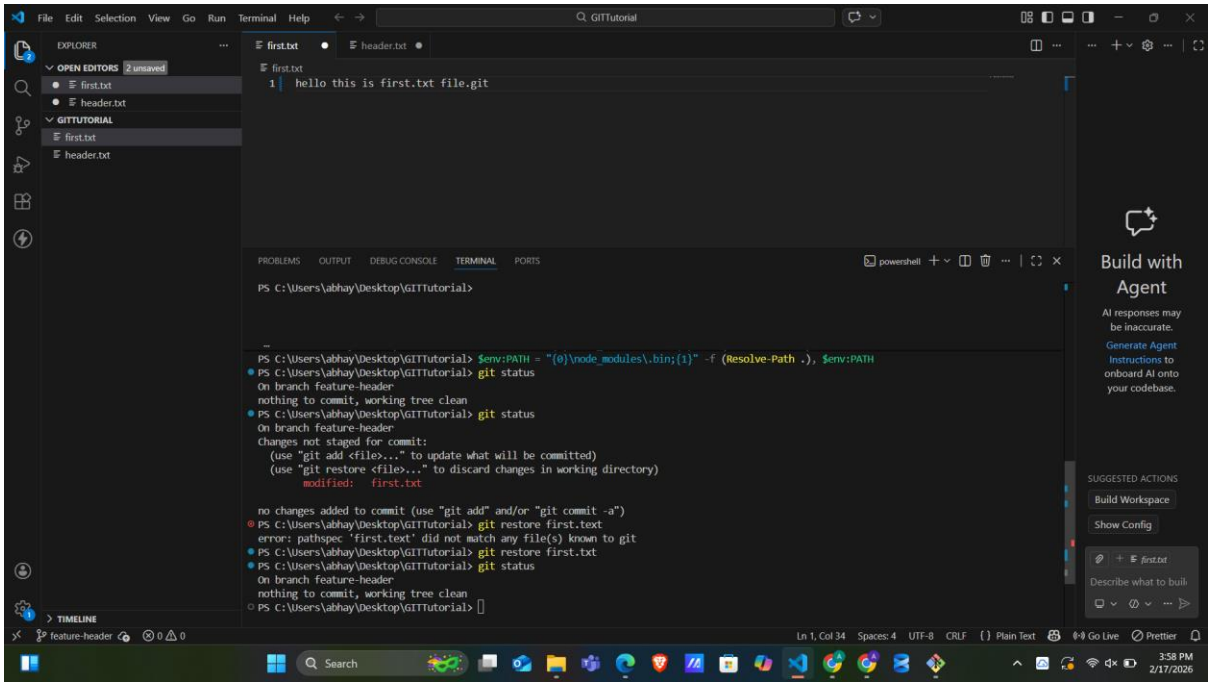
Steps:

1. In feature-header, modify index.txt

2. Check repository status

3. Undo the change using Git

4. Verify status again



TASK 5: Merge Feature Branch

Goal: Combine work into master

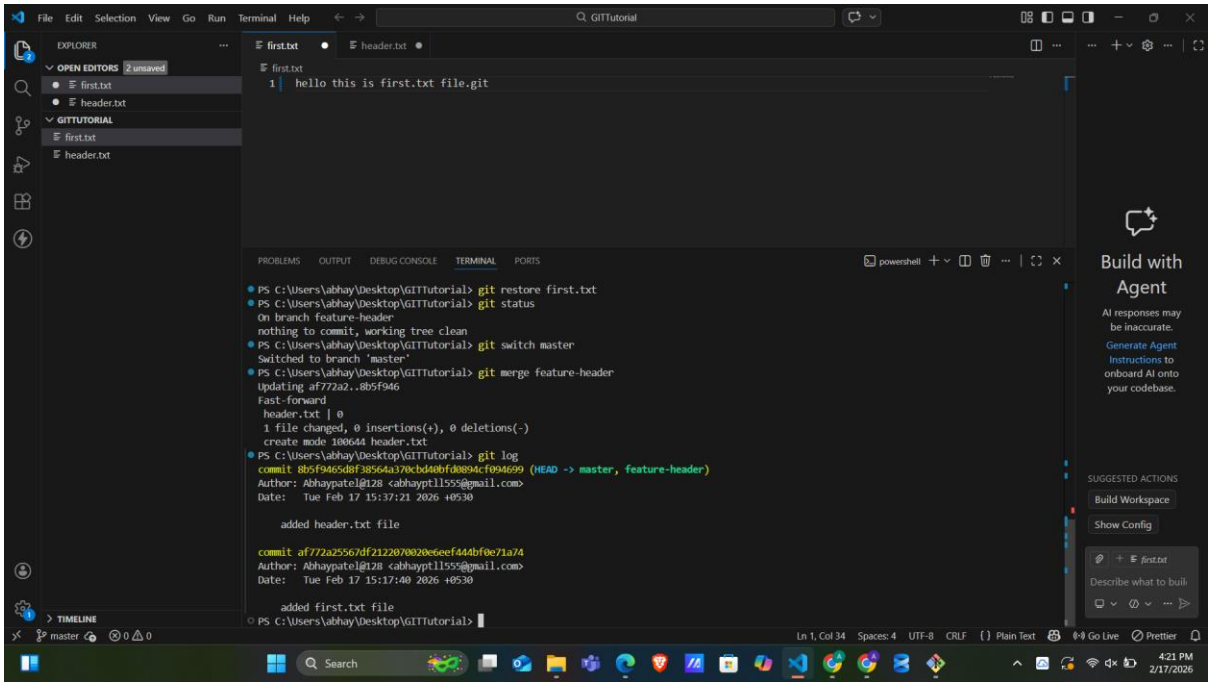
Steps:

1. Switch to master

2. Merge feature-header

3. Verify merged files

4. Check commit history



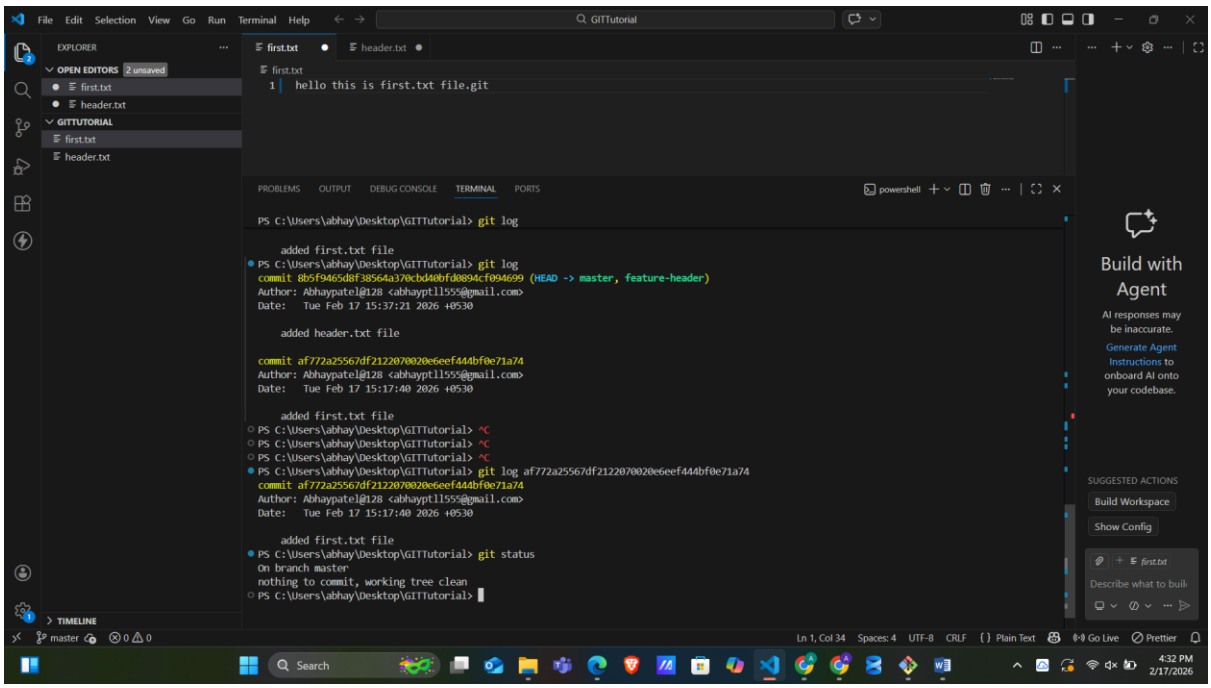
TASK 6: Checkout a Commit (Detached HEAD)

Goal: Understand commit-level navigation

Steps:

1. View commit history

2. Checkout any older commit

3. Check current branch status

4. Switch back safely to master